

Written Critique: Ethical Autonomous Vehicles

Kha Kein "Luke" Cheng

Kcheng314@gatech.edu

1 SCENARIO

Bicycle Scenario

2 REPORT QUESTIONS

2.1 Mercedes

The 2016 claim regarding Mercedes-Benz aligns with the Protectionist Ethics algorithm which prioritizes the vehicle's occupants first. If two autonomous cars operating under this identical algorithm, and neither car is programmed to give way or take the hit, they will end up crashing into each other.

But if these self-driving cars could communicate to one another, the outcome will change positively. For example, in the bicycle scenario allows both blue cars to work together and come up with a solution that works for both party rather than blindly executing the action that favors the occupants, such as minimizing the impact to sideswipe rather than a head on impact, thereby occupants in both cars get to survive.

As for as data privacy concern, this communication definitely introduces some issues as during the communication, each car needs to share precise GPS coordinates, speed, and necessary vehicle information. If this data is stored or unencrypted, it can easily leak or get misused such as people trying to hack the car using the information.

2.2 Tesla

This Tesla approach is closest to the Protectionist Ethics algorithm. The reason is it only focuses on the driver and ignores people outside the car. By refusing to take over the wheel even when the driver is doing something bad, it behaves with total loyalty to the driver and it is ok with anything as long as the driver is safe.

2.3 Moral Judgement

The car should adapt to its moral judgement automatically depending on the GPS location rather than letting human do it manually as human can be biased or forgetful. This adaptation is important as driving condition differs from country to country. For example, in the United States, roads are generally wider with very few bikers on the roads, and road condition is well-maintained. But in crowded city in country like India, the condition changes where the roads are much narrower and full of unpredictable mix of motorcycles, rickshaws and even pedestrians. So in this case, the car need to have a completely different setting where it may move at slower speed and do more screening and predictions of the surrounding.

2.4 Traffic Rules

The self-driving car would violate traffic law like staying in own lane or driving over the speed limit when implementing the Humanitarian algorithm where it swerves or speeds up in order to avoid a crash. It would also violate the same law when implementing the Protectionist algorithm when trying to save the vehicle occupants. As for the profit-based algorithms, it would intentionally hit whatever object that would result in a high insurance payout.

If all the traffic rules were followed, the individual that would be placed at harm would be the one who accidentally step into the car's legal lane. For example, in the bicycle scenario, the blue car ahead enters the street too slow, and the blue car behind tries to follow the rules and refuses to cross to the opposite lane where the cyclist is at to avoid collision, the only thing it can do is to apply hard on the brakes trying to slow down, if it cant stop in time, it will hit the car in front of it.